Subject: Fundamentals of Engineering Mathematics -II

Module 1 : Differential Equations of First Order and First Degree

1.1: Differential Equations in Growth and Decay; Formulation of Differential Equations

Differential Equations in Growth and Decay

Many natural, biological, and financial processes involve quantities that increase or decrease at a rate proportional to the quantity present. Such processes are modeled using first order differential equations.

Law of Natural Growth and Decay

The rate of change of a quantity at any time is **directly proportional** to the amount present at that time.

Mathematical Formulation

Let $y(t)$ be the quantity at time t. Then by the above statement,

$$\begin{aligned} \frac{dy}{dt} &\propto y \\ \Rightarrow \frac{dy}{dt} &= ky \qquad \cdots\cdots(*) \end{aligned}$$

where $k > 0 \Rightarrow$ Growth and $k < 0 \Rightarrow$ Decay

Solution of the Differential Equation

From (*) we get

$$\begin{aligned} \frac{dy}{dt} &= ky \\ \Rightarrow \frac{1}{y}dy &= k\ dt \\ \Rightarrow \ln|y| &= kt + \log C \\ \Rightarrow y &= Ce^{kt} \qquad \cdots\cdots(*) \end{aligned}$$

If we put $t = 0$ in (*) we get

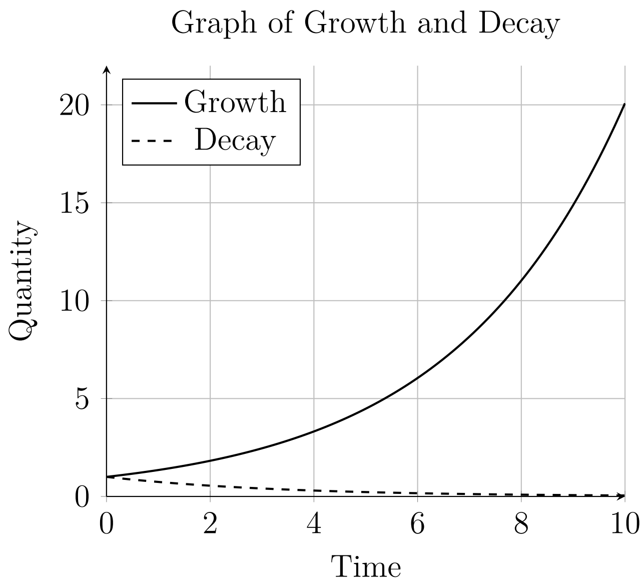
$$\begin{aligned} y = y(0) = y_0 &= Ce^{k(0)} \\ \Rightarrow y_0 &= C \end{aligned}$$

Hence

$$y = y_0 e^{kt}$$

where y_0 is the initial quantity is the **general solution** of growth and decay problems.

Growth Model: Here, the rate of **increase** of a quantity is proportional to the quantity present.
Decay Model: When the rate of **decrease** of a quantity is proportional to the quantity present.



Examples of Growth Model

- Population growth
- Bacterial growth
- Investment with continuous compounding

Model

$$y = y_0 e^{kt}, \quad k > 0$$

where y_0 is the initial quantity.
Example 1: Population Growth

Problem: The population of a town grows at a rate proportional to its population. If the population is 10,000 initially and increases by 20% in 10 years, formulate the differential equation and find the population after 20 years.
Solution: We have,

$$y = y_0 e^{kt}, \quad k > 0$$

Here, for $t = 10yrs,$, we have $y_0 = 10,000$ and

$$\begin{aligned}
y &= y_0 + 20\%(y_0) \\
\Rightarrow y &= = 10,000 + 2000 = 12000 \\
i.e,\ y &= y_0 e^{kt} = 12000 \\
i.e\ 12,000 &= 10,000 e^{10k} \\
\Rightarrow \frac{12,000}{10,000} &= e^{10k} \\
\Rightarrow \ln(1.2) &= \ln e^{10k} \\
\Rightarrow k &= \frac{\ln(1.2)}{10}
\end{aligned}$$

Therefore, the population after 20 years is given by

$$\begin{aligned}
y &= y_0 e^{kt} \\
\Rightarrow y &= 10,000 e^{\frac{\ln(1.2)}{10}(20)} \\
\Rightarrow y &= 10,000 e^{2\ln(1.2)} \\
\Rightarrow y &= 10,000(1.2)^2 = 14,400
\end{aligned}$$

Therefore, the population after 20 years is 14,400

Example 2: Financial Growth Example (Continuous Compounding)

An amount of ₹10000 is invested at 6% annual interest compounded continuously. Find:

1. Amount after 3 years
2. Time to double

Solution: $r = 0.06$, $A_0 = 10000$

$$A(t) = 10000e^{0.06t}$$

1) For Amount after 3 years

$$\begin{aligned}
A(3) &= 10000e^{0.18} \\
A(3) &\approx ₹\ 11972
\end{aligned}$$

2) To find time (t) when amount becomes double (₹20000)

$$\begin{aligned}
10000e^{0.06t} &= 20000 \\
e^{0.06t} &= 2 \\
0.06t &= \ln 2 \\
t &= \frac{\ln 2}{0.06} \\
t &\approx 11.55 \text{ years}
\end{aligned}$$

Problem: An investment of ₹ 50,000 grows at a continuous rate of 6% per annum. Find the amount after 5 years.
Solution:

$$A = A_0 e^{rt}$$

$$A = 50000e^{0.06\times 5}$$

$$\boxed{A \approx ₹67,493}$$

Examples Decay Model

- Radioactive decay
- Cooling of objects
- Depreciation of assets

Model

$$\begin{aligned} y &= y_0 e^{-kt}, \quad k > 0 \\ or \ y &= y_0 e^{kt}, \quad k < 0 \end{aligned}$$

where y_0 is the initial quantity.
Example 3: Radioactive Decay
A radioactive substance decays at a rate proportional to the amount present. If 10% of the substance decays in 5 years, find the amount left after 15 years.
Solution: We have,

$$y = y_0 e^{-kt}, \quad k > 0$$

Here, for $t = 5yrs,$, we have $y_0 = y_0$ and

$$\begin{aligned} y = y(5) &= y_0 e^{-k(5)} \\ Given: \ y(5) &= y_0 - 10\% y_0 = 0.9y_0 \\ \Rightarrow y_0 e^{-k(5)} &= 0.9y_0 \\ \Rightarrow -0.5k &= \ln(0.9) \\ \Rightarrow k &= -\frac{\ln(0.9)}{5} = 0.02107 \\ \Rightarrow y &= y_0 e^{-0.02107t} \end{aligned}$$

Therefore, the the amount left after 15 years is given by

$$\begin{aligned} y = y(15) &= y_0 e^{-k(15)} \\ \Rightarrow y(15) &= y_0 e^{-0.02107(15)} = 0.729y_0 \end{aligned}$$

Example 4: D.E of Circles
Find the differential equation of the family of all circles, which pass through the origin and whose centre lies on y - axis.
Solution: The general equations of family of circles, which pass through the origin and whose centre lies on y - axis

$$(x-0)^2+(y-a)^2=a^2 \implies x^2+y^2-2ay+a^2=a^2$$
$$\implies x^2+y^2-2ay=0$$

where a is an arbitrary radius of the circle.
Differentiating the equation of circle

$$2x+2y\frac{dy}{dx}-2a\frac{dy}{dx}=0 \implies x+y\frac{dy}{dx}-a\frac{dy}{dx}=0$$

Eliminating the arbitrary constant a and substitute in the above differential equation:
We have $x^2+y^2-2ay=0 \implies a=\dfrac{x^2+y^2}{2y}$
and

$$x+y\frac{dy}{dx}-\left(\frac{x^2+y^2}{2y}\right)\frac{dy}{dx}=0 \implies 2xy=-2y^2\frac{dy}{dx}+\left(x^2+y^2\right)\frac{dy}{dx}$$

$$2xy=\left(-2y^2+x^2+y^2\right)\frac{dy}{dx} \implies \left(x^2-y^2\right)\frac{dy}{dx}=2xy$$

Thus, the required differential equation is: $\left(x^2-y^2\right)\dfrac{dy}{dx}=2xy$.
Key Points

- Growth and decay problems lead to separable differential equations
- Solutions are exponential in nature
- Always identify whether the problem represents growth or decay
- Initial conditions are used to determine constants

Practice Problems in Formulation

1. (Half Life of Uranium) Uranium is an alternate source of fuel for the generation of electricity. It exhibits the following property: It disintegrates at a rate proportional to the amount present at any given instant.
   If m_1 and m_2 g of Uranium are present in the furnace at time t_1 and t_2 respectively, find the half life of Uranium

2. (Antenna Design) A reflector reflects signals in parallel, when these signals are coming from a point source. A receiver receives these parrallel signals and converges to a point when these signals are coming from a long distance. Design a shape of the reflector /receiver for the antenna in a communication system

3. (Escape Velocity) A satellite is launched via a rocket to a particular orbit. Find the initial velocity of the satellite which is fired via a multi-stage rocket in radial direction from the earth's centre and is supposed to escape the earth.

4. (Newton's Law of Cooling) Newton's Law of Cooling states that the rate at which a substance cools in moving air is proportional to the difference between the temperature of the substance and the temperature of the air. Formulate the differential equation and obtain the temperature T of a substance at any time t.

5. ($L - R$ circuit) A resistance R and an inductance L are connected in a series with a voltage supply $E(t)$. Find the current in the circuit.

6. ($R - C$ circuit) A resistance R and a Capacitor C are connected in a series with a voltage supply $E(t)$. Find the current in the circuit.

7. A person deposits ₹1000 in a bank with 10% compound interest over 5 years. What is the return value after 5 years?

8. A population grows in a country at the rate of r_0 per year. How long does it take for the population to double?